

Algoritmo		Handshake		Pacotes Troca	Tamanho
Troca de Chave	Assinatura	Latência [ms]	Amplificação Realizada	Número	Tamanho
BP256 + ML-KEM-512	P256 + ML-DSA-44	7.6±0.5	3.10x	5 / 9	20
BP256 + ML-KEM-512	RSA3072 + ML-DSA-44	9.0±0.4	3.49x	5 / 10	20
BP256 + ML-KEM-512	P256 + FALCON-512	7.7±0.3	1.55x	4 / 6	20
BP256 + ML-KEM-512	P256 + FALCON-PADDED-512	7.7±0.4	1.56x	4 / 6	20
BP256 + ML-KEM-512	RSA3072 + FALCON-512	8.9±0.3	1.94x	4 / 7	20
BP256 + ML-KEM-512	RSA3072 + FALCON-PADDED-512	9.7±0.8	1.95x	4 / 7	20
BP256 + ML-KEM-512	P256 + SPHINCS-SHA2-128-F-SIMPLE	13.7±0.5	13.73x	8 / 33	20
BP256 + ML-KEM-512	P256 + SPHINCS-SHA2-128-S-SIMPLE	125.4±1.2	6.85x	6 / 17	20
BP256 + ML-KEM-512	RSA3072 + SPHINCS-SHA2-128-F-SIMPLE	15.1±0.3	13.49x	11 / 34	20
BP256 + ML-KEM-512	RSA3072 + SPHINCS-SHA2-128-S-SIMPLE	126.0±0.6	7.23x	6 / 18	20
BP256 + ML-KEM-512	P256 + SPHINCS-SHAKE-128-F-SIMPLE	21.6±1.1	13.73x	8 / 33	20
BP256 + ML-KEM-512	P256 + SPHINCS-SHAKE-128-S-SIMPLE	261.3±3.1	6.84x	6 / 17	20
BP256 + ML-KEM-512	RSA3072 + SPHINCS-SHAKE-128-F-SIMPLE	22.3±0.3	14.11x	8 / 34	20
BP256 + ML-KEM-512	RSA3072 + SPHINCS-SHAKE-128-S-SIMPLE	263.3±1.5	7.23x	6 / 18	20
P256 + ML-KEM-512	P256 + ML-DSA-44	6.5±0.3	3.10x	5 / 9	20
P256 + ML-KEM-512	RSA3072 + ML-DSA-44	7.7±0.2	3.49x	5 / 10	20
P256 + ML-KEM-512	P256 + FALCON-512	6.5±0.2	1.55x	4 / 6	20
P256 + ML-KEM-512	P256 + FALCON-PADDED-512	6.5±0.2	1.56x	4 / 6	20
P256 + ML-KEM-512	RSA3072 + FALCON-512	7.8±0.2	1.94x	4 / 7	20
P256 + ML-KEM-512	RSA3072 + FALCON-PADDED-512	8.1±0.3	1.95x	4 / 7	20
P256 + ML-KEM-512	P256 + SPHINCS-SHA2-128-F-SIMPLE	12.3±0.1	13.33x	10 / 33	20
P256 + ML-KEM-512	P256 + SPHINCS-SHA2-128-S-SIMPLE	122.9±0.1	6.85x	6 / 17	20
P256 + ML-KEM-512	RSA3072 + SPHINCS-SHA2-128-F-SIMPLE	13.9±0.2	14.11x	8 / 34	20
P256 + ML-KEM-512	RSA3072 + SPHINCS-SHA2-128-S-SIMPLE	124.3±0.2	7.23x	6 / 18	20
P256 + ML-KEM-512	P256 + SPHINCS-SHAKE-128-F-SIMPLE	19.4±0.2	13.73x	8 / 33	20
P256 + ML-KEM-512	P256 + SPHINCS-SHAKE-128-S-SIMPLE	257.4±0.3	6.84x	6 / 17	20
P256 + ML-KEM-512	RSA3072 + SPHINCS-SHAKE-128-F-SIMPLE	20.8±0.1	14.11x	8 / 34	20
P256 + ML-KEM-512	RSA3072 + SPHINCS-SHAKE-128-S-SIMPLE	259.1±0.5	7.23x	6 / 18	20
P256 + BIKE-L1	P256 + ML-DSA-44	7.3±0.3	2.34x	6 / 10	30
P256 + BIKE-L1	RSA3072 + ML-DSA-44	8.7±0.4	2.61x	6 / 11	30
P256 + BIKE-L1	P256 + FALCON-512	7.1±0.3	1.30x	4 / 6	30
P256 + BIKE-L1	P256 + FALCON-PADDED-512	7.3±0.3	1.31x	4 / 6	30
P256 + BIKE-L1	RSA3072 + FALCON-512	8.6±0.2	1.58x	4 / 7	30
P256 + BIKE-L1	RSA3072 + FALCON-PADDED-512	8.8±0.2	1.58x	4 / 7	30
P256 + BIKE-L1	P256 + SPHINCS-SHA2-128-F-SIMPLE	13.3±0.3	9.83x	8 / 34	30
P256 + BIKE-L1	P256 + SPHINCS-SHA2-128-S-SIMPLE	124.0±0.3	4.97x	6 / 18	30
P256 + BIKE-L1	RSA3072 + SPHINCS-SHA2-128-F-SIMPLE	14.7±0.1	9.98x	9 / 35	30
P256 + BIKE-L1	RSA3072 + SPHINCS-SHA2-128-S-SIMPLE	125.2±0.2	5.18x	7 / 19	30
P256 + BIKE-L1	P256 + SPHINCS-SHAKE-128-F-SIMPLE	20.2±0.2	9.83x	8 / 34	30
P256 + BIKE-L1	P256 + SPHINCS-SHAKE-128-S-SIMPLE	258.1±0.2	4.97x	6 / 18	30
P256 + BIKE-L1	RSA3072 + SPHINCS-SHAKE-128-F-SIMPLE	21.5±0.1	10.09x	8 / 35	30
P256 + BIKE-L1	RSA3072 + SPHINCS-SHAKE-128-S-SIMPLE	259.4±0.3	5.24x	6 / 19	30
P256 + FRODO-640-AES	P256 + ML-DSA-44	7.2±0.2	1.38x	14 / 19	120
P256 + FRODO-640-AES	RSA3072 + ML-DSA-44	8.9±0.2	1.50x	13 / 18	120
P256 + FRODO-640-AES	P256 + FALCON-512	7.2±0.2	1.08x	14 / 15	130
P256 + FRODO-640-AES	P256 + FALCON-PADDED-512	7.3±0.2	1.09x	12 / 13	120
P256 + FRODO-640-AES	RSA3072 + FALCON-512	8.8±0.2	1.15x	14 / 16	130
P256 + FRODO-640-AES	RSA3072 + FALCON-PADDED-512	8.9±0.3	1.15x	14 / 16	130
P256 + FRODO-640-AES	P256 + SPHINCS-SHA2-128-F-SIMPLE	13.4±0.3	3.46x	15 / 41	130
P256 + FRODO-640-AES	P256 + SPHINCS-SHA2-128-S-SIMPLE	124.1±0.5	2.23x	14 / 25	120
P256 + FRODO-640-AES	RSA3072 + SPHINCS-SHA2-128-F-SIMPLE	14.8±0.2	3.84x	17 / 42	120
P256 + FRODO-640-AES	RSA3072 + SPHINCS-SHA2-128-S-SIMPLE	125.6±0.5	2.32x	13 / 26	120
P256 + FRODO-640-AES	P256 + SPHINCS-SHAKE-128-F-SIMPLE	20.1±0.1	3.77x	16 / 41	120
P256 + FRODO-640-AES	P256 + SPHINCS-SHAKE-128-S-SIMPLE	258.5±0.5	2.24x	13 / 25	120
P256 + FRODO-640-AES	RSA3072 + SPHINCS-SHAKE-128-F-SIMPLE	21.9±0.3	3.84x	17 / 42	120
P256 + FRODO-640-AES	RSA3072 + SPHINCS-SHAKE-128-S-SIMPLE	260.6±1.1	2.32x	13 / 26	120
P256 + FRODO-640-SHAKE	P256 + ML-DSA-44	8.8±0.4	1.39x	13 / 19	130
P256 + FRODO-640-SHAKE	RSA3072 + ML-DSA-44	10.4±0.5	1.50x	14 / 18	120
P256 + FRODO-640-SHAKE	P256 + FALCON-512	9.0±0.3	1.08x	13 / 15	130
P256 + FRODO-640-SHAKE	P256 + FALCON-PADDED-512	9.0±0.3	1.09x	12 / 13	120
P256 + FRODO-640-SHAKE	RSA3072 + FALCON-512	10.5±0.3	1.15x	13 / 16	130
P256 + FRODO-640-SHAKE	RSA3072 + FALCON-PADDED-512	10.5±0.2	1.16x	13 / 16	130
P256 + FRODO-640-SHAKE	P256 + SPHINCS-SHA2-128-F-SIMPLE	15.1±0.4	3.46x	15 / 41	130
P256 + FRODO-640-SHAKE	P256 + SPHINCS-SHA2-128-S-SIMPLE	125.3±0.1	2.23x	14 / 25	120